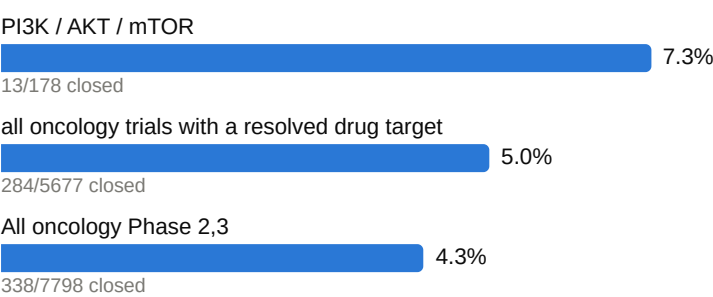


PI3K / AKT / mTOR: 9 molecules stopped early, all small molecules

13 stopped trials, but 9 molecules — all small molecules: Copanlisib, Taselisib, Vistusertib, Capivasertib and 5 more. 1 of the stops were the trial's own verdict and 4 followed a decision taken elsewhere; 8 cannot be established. Against 5.0% for all oncology trials with a resolved drug target and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

7.3%	9 / 3 / 1	50.6%	2.1%
13 of 178 closed trials (95% CI 4.3%–12.1%)	efficacy / safety / both (adds to 13)	of 352 trials have closed · 174 still open or unresolved	stopped for a biological reason within 3 years of starting (95% CI 1.0%–4.7%; 199 still at risk; 1.5% area-wide)

How this compares



What is in this class: 253 distinct experimental drugs across 352 trials (small molecule 351, antibody 94, unknown 19, adc 14). Most tested: ALPELISIB (41), COPANLISIB (38), IPATASERTIB (33), PACLITAXEL (26), FULVESTRANT (24). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 13 stops came from 11 sponsor-asset programmes across 10 sponsors; the largest (Samsung Medical Center) contributed 2; removing that programme's trials from both sides leaves 6.3%. Screened alongside 83 other segments in this area, it does not survive a 10% false-discovery correction (q=1, Benjamini-Yekutieli). A further 19 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 18.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	10	90	11.1%	6.1%–19.3%
2018–2020	3	56	5.4%	1.8%–14.6%
2021–2024	0	32	0.0%	0.0%–10.7%

Where the stops sit

Efficacy / futility: 9, Safety: 3, Efficacy and safety: 1. Sponsors with most stops: Samsung Medical Center (3), Bayer (2), Roche (1), Fondazione Italiana Linfomi - ETS (1), National Heart, Lung, and Blood Institute (NHLBI) (1), UNICANCER (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02626455	3	Bayer	COPANLISIB	2023-09	Efficacy	Company decided to terminate this study due to study did not meet the primary endpoint.
NCT03458728	2	Bayer	COPANLISIB	2023-02	Efficacy	no anticipated benefit over available standard therapies
NCT02340221	3	Roche	TASELISIB	2021-06	Efficacy + safety	The Sponsor discontinued the manufacturing and development of taselisib due to modest clinical benefit and limited tolerability.
NCT03890289	2	Fondazione Italiana Linfomi - ETS	IDELALISIB, OBINUTUZUMAB	2021-04	Safety	Study was halted prematurely due to safety, since toxicity stopping rules have been met

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04209621	2	National Heart, Lung, and Blood Institute (NHLBI)	DUVELISIB, IBRUTINIB	2021-01	Safety	Due to unexpected sudden death on study.
NCT02822482	2	UNICANCER	COPANLISIB, CETUXIMAB	2019-10	Efficacy	futility of the treatment

The 6 most recent of 13 stopped trials. Every molecule behind all 13 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated).
Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 50.6% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

Any mechanism, sponsor or indication, updated weekly. The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: clinicaltrialfailures.com/data-licensing

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.